Advanced Trigonometry

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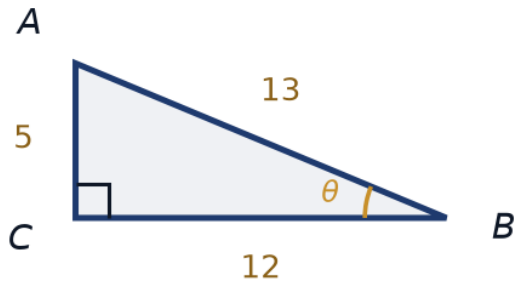
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Compound angle formulas

Expand $\sin(x + 30^\circ)$ in terms of $\sin x$ and $\cos x$, using exact values for $\sin 30^\circ$ and $\cos 30^\circ$.

2 [Foundational] • Target: Compound angle formulas

Given $\sin A = \frac{3}{5}$ and $\cos B = \frac{12}{13}$, with A and B both acute, find $\cos(A + B)$.



3 [Foundational] • Target: Compound angle formulas

Use the compound-angle formula for tangent to find the exact value of $\tan 75^\circ$, using $75^\circ = 45^\circ + 30^\circ$.

4 [Foundational] • Target: Compound angle formulas

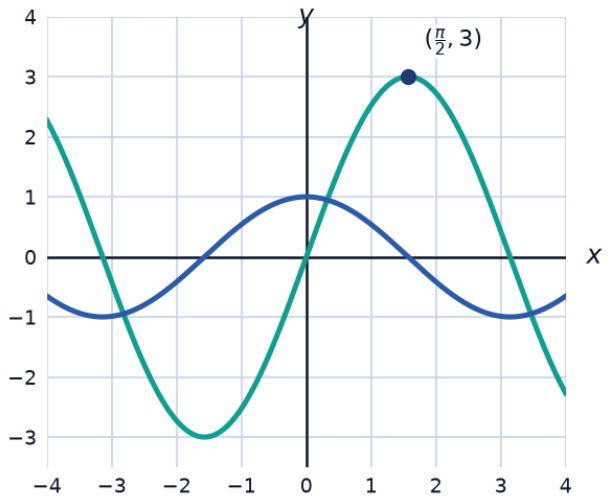
Use the compound-angle formula for cosine to find the exact value of $\cos 15^\circ$, using $15^\circ = 45^\circ - 30^\circ$.

5 [Foundational] • Target: Double angle formulas

Given $\sin\theta = \frac{4}{5}$ with θ acute, find $\sin 2\theta$ and $\cos 2\theta$.

6 [Foundational] • Target: Double angle formulas

Given $\cos\theta = -\frac{1}{3}$ with $90^\circ < \theta < 180^\circ$, find $\cos 2\theta$ and state which quadrant 2θ lies in.



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7 [Foundational] • Target: Double angle formulas

Given $\tan\theta = \frac{1}{2}$ with θ acute, find $\tan 2\theta$, using $\tan 2\theta = \frac{2\tan\theta}{1-\tan^2\theta}$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Solving equations using double angle formulas

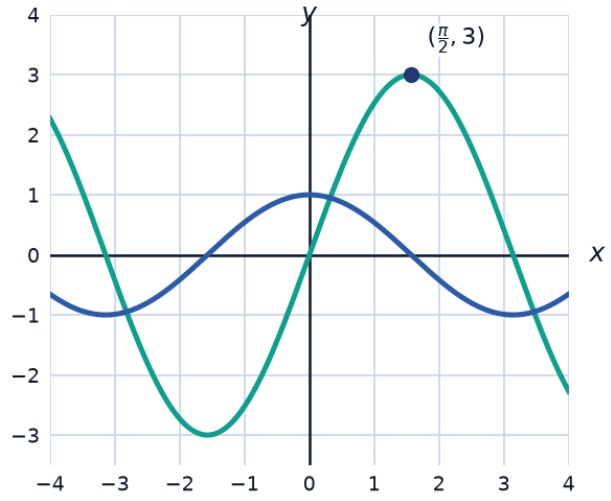
Solve $\cos 2\theta = \sin\theta$ for $0^\circ \le \theta < 360^\circ$, using $\cos 2\theta = 1 - 2\sin^2\theta$.

9 [Intermediate] • Target: Solving equations using double angle formulas

Solve $\sin 2\theta = \cos\theta$ for $0^\circ \le \theta < 360^\circ$, using $\sin 2\theta = 2\sin\theta\cos\theta$.

10 [Intermediate] • Target: The R sin(θ+α) form

Express $3\sin\theta + 4\cos\theta$ in the form $R\sin(\theta+\alpha)$, with $R > 0$ and $0^\circ < \alpha < 90^\circ$, giving α to the nearest degree.



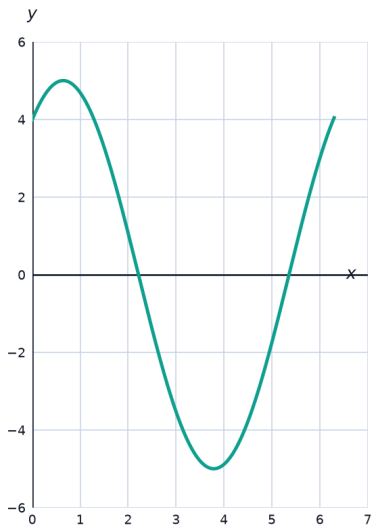
11 [Intermediate] • Target: The R sin(θ+α) form

Express $\sqrt{3}\sin\theta - \cos\theta$ in the form $R\sin(\theta-\alpha)$, giving exact values of R and α.

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12 [Intermediate] • Target: Using the R-form to find extrema and solve equations

The graph shows $y = 5\sin(x + 53^\circ)$, the R-form of $3\sin x + 4\cos x$ found earlier. State the maximum and minimum values of $3\sin\theta + 4\cos\theta$, and the value of θ (to the nearest degree, $0^\circ \le \theta < 360^\circ$) at which the maximum occurs.



13 [Intermediate] • Target: Using the R-form to find extrema and solve equations

Solve $\sqrt{3}\sin\theta - \cos\theta = 1$ for $0^\circ \le \theta < 360^\circ$, using the R-form $2\sin(\theta - 30^\circ)$ found earlier.

14 [Intermediate] • Target: Proving trigonometric identities

Prove that $\frac{\sin 2\theta}{1 + \cos 2\theta} = \tan\theta$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

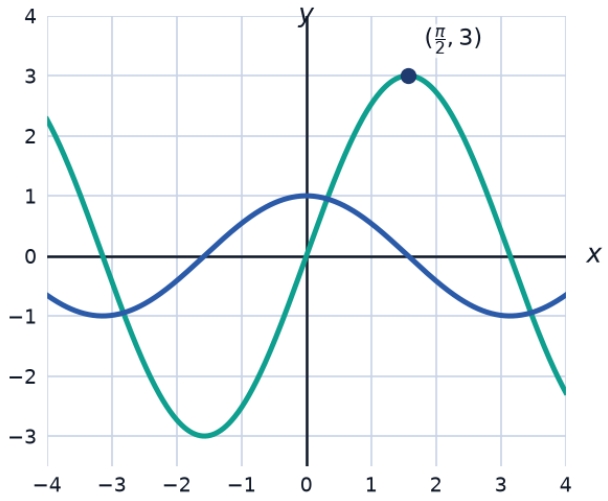
15 [Challenging] • Target: Proving trigonometric identities • Cross-Lesson: 4.01 Differentiation

Prove that $\cos^4\theta - \sin^4\theta = \cos 2\theta$.

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16 [Challenging] • Target: Proving trigonometric identities • Cross-Lesson: 5.01 Integration

Prove that $(\sin\theta + \cos\theta)^2 = 1 + \sin 2\theta$.



17 [Challenging] • Target: Solving equations that combine several identities • Cross-Lesson: 6.01 Sequences & Series

Solve $2\cos 2\theta + 3\sin\theta = 1$ for $0^\circ \le \theta < 360^\circ$, using $\cos 2\theta = 1 - 2\sin^2\theta$, giving answers to one decimal place where needed.

18 [Challenging] • Target: Solving equations that combine several identities • Cross-Lesson: 7.01 Vectors

Solve $\cos 2\theta = 1 - \sin\theta$ for $0^\circ \le \theta < 360^\circ$, using $\cos 2\theta = 1 - 2\sin^2\theta$.

19 [Challenging] • Target: A well-designed word problem: alternating current • Cross-Lesson: 8.01 Algebra: Partial Fractions & Division

This question has three parts. The voltage in an AC circuit is modeled by $V(t) = 6\sin(100\pi t) + 8\cos(100\pi t)$ volts, where t is time in seconds. (a) Express $V(t)$ in the form $R\sin(100\pi t + \alpha)$, giving R exactly and α in radians to four decimal places. (b) State the maximum voltage reached, and the smallest positive value of t (in seconds, to four significant figures) at which it occurs. (c) Find the two values of t in $0 \le t < 0.02$ at which $V(t) = 5$, to four significant figures.

20 [Challenging] • Target: A well-designed word problem: alternating current • Cross-Lesson: 9.01 Complex Numbers

Express $4\sin\theta + 3\cos\theta$ in the form $R\sin(\theta \pm \alpha)$, $R > 0$. Find R.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Reciprocal & Pythagorean Identities

Solve the equation $2\tan^2\theta + \sec\theta - 1 = 0$ for $0^\circ \leq \theta \leq 360^\circ$.

a. $\theta = 60^\circ, 180^\circ, 300^\circ$

b. $\theta = 60^\circ, 300^\circ$

c. $\theta = 120^\circ, 240^\circ$

d. $\theta = 45^\circ, 135^\circ$

22 [Foundational] • Target: Reciprocal & Pythagorean Identities

Simplify $\frac{1}{\sec\theta - \tan\theta}$.

a. $\sec\theta + \tan\theta$

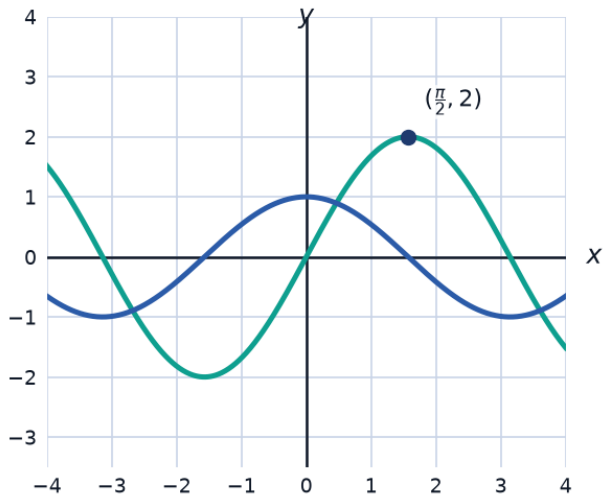
b. $\sec\theta - \tan\theta$

c. $\cos\theta + \sin\theta$

d. 1

23 [Foundational] • Target: Reciprocal & Pythagorean Identities

Given $\cot\theta = 2$ and θ is in Quadrant III, find the exact value of $\csc\theta$.



a. $-\sqrt{5}$

b. $\sqrt{5}$

c. $-\frac{\sqrt{5}}{2}$

d. $-\frac{1}{2}$

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24 [Foundational] • Target: Compound Angle Formulas

Expand $\sin(A+B)$ using the addition formula.

a. $\sin A\cos B+\cos A\sin B$

b. $\sin A\cos B-\cos A\sin B$

c. $\cos A\cos B-\sin A\sin B$

d. $\sin A\sin B+\cos A\cos B$

25 [Foundational] • Target: Compound Angle Formulas

Find the exact value of $\tan(75^\circ)$ using compound angle identities.

a. $2+\sqrt{3}$

b. $2-\sqrt{3}$

c. $\sqrt{3}+1$

d. $\frac{\sqrt{3}+1}{\sqrt{3}-1}$

26 [Foundational] • Target: Compound Angle Formulas

If $\tan A=2$ and $\tan B=3$, find the exact value of $\tan(A+B)$.

a. -1

b. 1

c. 5

d. -5

27 [Foundational] • Target: Double Angle Formulas

Express $\cos(2\theta)$ strictly in terms of $\sin\theta$.

a. $1-2\sin^2\theta$

b. $2\sin^2\theta-1$

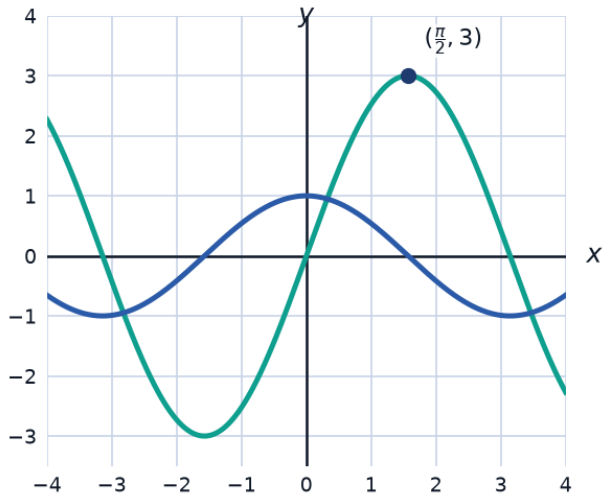
c. $1-\sin^2\theta$

d. $2\cos^2\theta-1$

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28 [Foundational] • Target: Double Angle Formulas

Solve the equation $\sin(2x) = \sin(x)$ for $0 \le x \le 2\pi$.



a. $x = 0, \pi/3, \pi, 5\pi/3, 2\pi$

b. $x = 0, \pi, 2\pi$

c. $x = \pi/3, 5\pi/3$

d. $x = 0, \pi/2, \pi, 3\pi/2, 2\pi$

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Double Angle Formulas

Prove that $\frac{\sin(2\theta)}{1+\cos(2\theta)} \equiv \tan\theta$. What is the denominator after substitution?

a. $2\cos^2\theta$

b. $2\sin^2\theta$

c. $1 - \cos^2\theta$

d. $2\cos\theta$

30 [Intermediate] • Target: The $R\cos(\theta \pm \alpha)$ / $R\sin(\theta \pm \alpha)$ Method

Express $3\cos\theta + 4\sin\theta$ in the form $R\cos(\theta - \alpha)$ where $R > 0$ and $0^\circ < \alpha < 90^\circ$.

a. $5\cos(\theta - 53.1^\circ)$

b. $5\cos(\theta - 36.9^\circ)$

c. $7\cos(\theta - 53.1^\circ)$

d. $5\cos(\theta + 53.1^\circ)$

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31 [Intermediate] • Target: The $R\cos(\theta \pm \alpha)$ / $R\sin(\theta \pm \alpha)$ Method

Find the maximum value of the function $f(\theta) = 5\cos\theta - 12\sin\theta + 7$ and the smallest positive θ at which it occurs.

a. Maximum is 20

b. Maximum is 13

c. Maximum is 26

d. Maximum is 19

32 [Intermediate] • Target: The $R\cos(\theta \pm \alpha)$ / $R\sin(\theta \pm \alpha)$ Method

Solve $3\cos x + 4\sin x = 2.5$ for $0^\circ \le x \le 360^\circ$.

a. $x = 113.1^\circ$ and $x = 353.1^\circ$

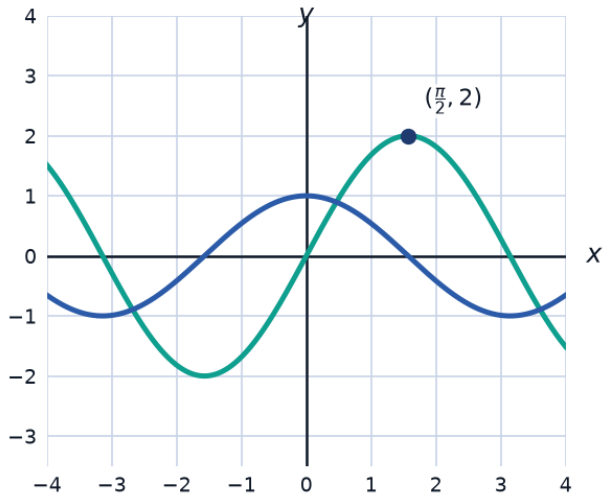
b. $x = 60^\circ$ and $x = 300^\circ$

c. $x = 90^\circ$ and $x = 270^\circ$

d. $x = 45^\circ$ and $x = 225^\circ$

33 [Intermediate] • Target: Compound angle formulas

Express $4\sin\theta - 3\cos\theta$ in the form $R\sin(\theta \pm \alpha)$, $R > 0$. Find R.



a. 7

b. 5

c. 6

d. 1

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34 [Intermediate] • Target: Double angle formulas

Express $216\sin\theta - 195\cos\theta$ in the form $R\sin(\theta \pm \alpha)$, $R > 0$. Find R.

a. 21

b. 294

c. 411

d. 291

35 [Intermediate] • Target: Solving equations using double angle formulas

Express $40\sin\theta - 42\cos\theta$ in the form $R\sin(\theta \pm \alpha)$, $R > 0$. Find R.

a. 2

b. 82

c. 60

d. 58

36 [Intermediate] • Target: The R sin(θ+α) form

Express $33\sin\theta - 56\cos\theta$ in the form $R\sin(\theta \pm \alpha)$, $R > 0$. Find R.

a. 23

b. 66

c. 89

d. 65

37 [Intermediate] • Target: Using the R-form to find extrema and solve equations

Express $84\sin\theta - 13\cos\theta$ in the form $R\sin(\theta \pm \alpha)$, $R > 0$. Find R.

a. 97

b. 86

c. 85

d. 71

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Proving trigonometric identities • Cross-Lesson: 7.01 Vectors

Express $21\sin\theta + 20\cos\theta$ in the form $R\sin(\theta \pm \alpha)$, $R > 0$. Find R.

a. 1

b. 41

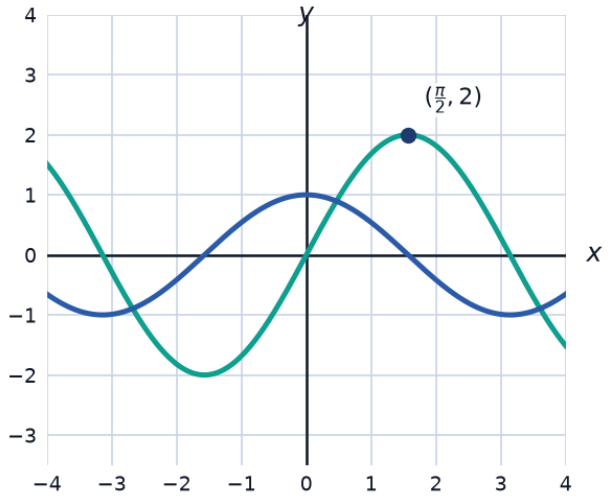
c. 29

d. 30

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39 [Challenging] • Target: Solving equations that combine several identities • Cross-Lesson: 8.01 Algebra: Partial Fractions & Division

Express $21\sin\theta - 20\cos\theta$ in the form $R\sin(\theta \pm \alpha)$, $R > 0$. Find R.



a. 1

b. 41

c. 30

d. 29

40 [Challenging] • Target: A well-designed word problem: alternating current • Cross-Lesson: 9.01 Complex Numbers

Express $9\sin\theta - 40\cos\theta$ in the form $R\sin(\theta \pm \alpha)$, $R > 0$. Find R.

a. 41

b. 31

c. 42

d. 49

41 [Challenging] • Target: Reciprocal & Pythagorean Identities • Cross-Lesson: 1.01 Functions: Modulus & Inverse

Express $24\sin\theta - 10\cos\theta$ in the form $R\sin(\theta \pm \alpha)$, $R > 0$. Find R.

a. 28

b. 26

c. 14

d. 34

42 [Challenging] • Target: Compound Angle Formulas • Cross-Lesson: 10.01 Numerical Methods

Express $5\sin\theta + 12\cos\theta$ in the form $R\sin(\theta \pm \alpha)$, $R > 0$. Find R.

a. 7

b. 13

c. 17

d. 14

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43 [Challenging] • Target: Double Angle Formulas • Cross-Lesson: 11.01 Differential Equations

Express $15\sin\theta + 8\cos\theta$ in the form $R\sin(\theta \pm \alpha)$, $R > 0$. Find R.

a. 23

b. 7

c. 17

d. 18

44 [Challenging] • Target: The $R\cos(\theta \pm \alpha)$ / $R\sin(\theta \pm \alpha)$ Method • Cross-Lesson: 12.01 Poisson Distribution

Express $8\sin\theta + 15\cos\theta$ in the form $R\sin(\theta \pm \alpha)$, $R > 0$. Find R.

a. 17

b. 18

c. 7

d. 23

45 [Challenging] • Target: Compound angle formulas • Cross-Lesson: 13.01 Continuous Random Variables

Given $120\sin\theta + 22\cos\theta = R\sin(\theta \pm \alpha)$, $R > 0$, find R^2.

a. 15004

b. 14884

c. 14883

d. 20164

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